



# Digital Signature Scheme



# RSA Digital Signature Scheme

- RSA can also be used to **sign and verify messages**.

- In Digital Signature:

- Sender's private key  $\rightarrow$  used for signing
- Sender's public key  $\rightarrow$  used for verification

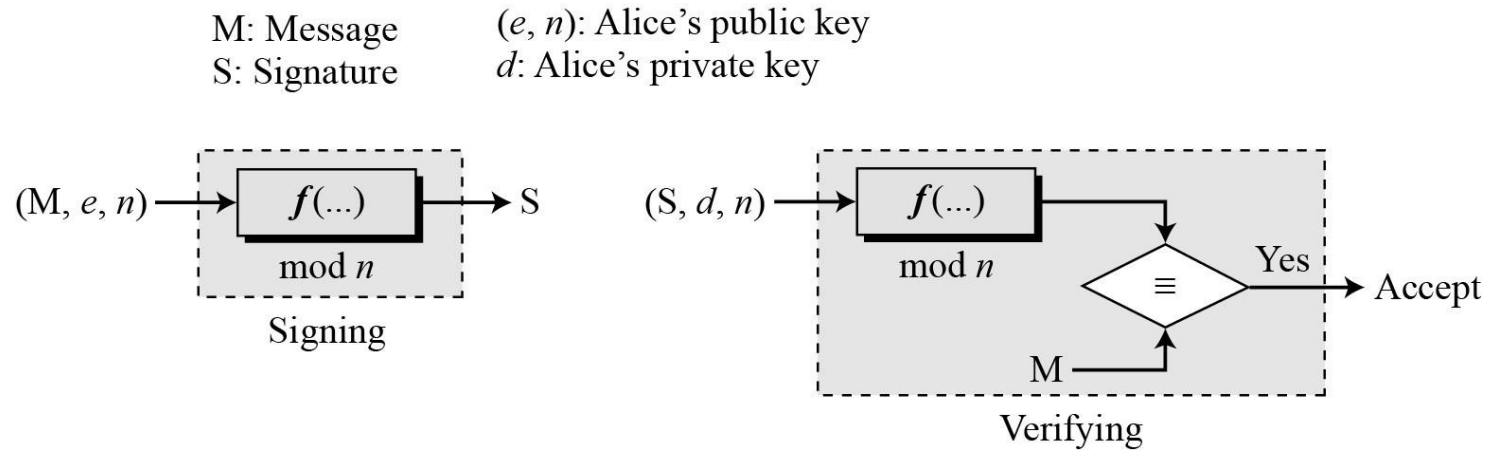
## Steps:

1. Sender creates signature  
 **$S = \text{Sign}(M, d)$**  using private key
2. Sender sends  
**Message (M) + Signature (S)**
3. Receiver verifies signature using sender's public key
4. If verified message = original message

✓ Accept  
Otherwise  
✗ Reject

## This ensures:

- **Authentication** (sender identity)
- **Integrity** (message not changed)



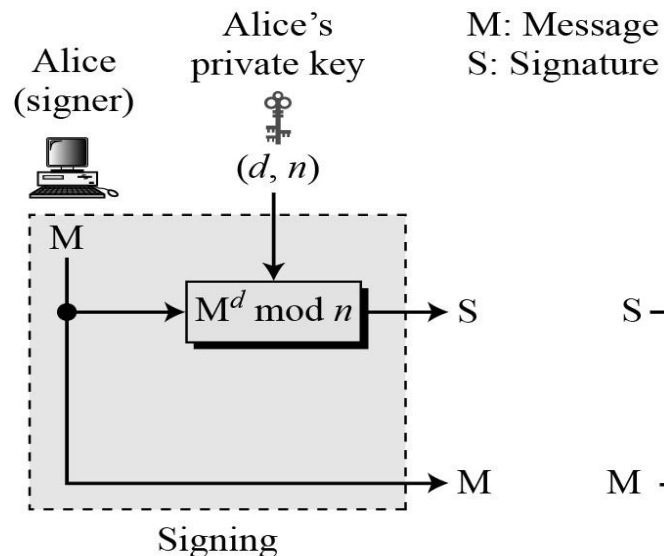
# RSA Digital Signature Scheme

## Signing (Sender – Alice)

1. Alice takes message **M**
2. Uses her **private key (d, n)**
3. Generates signature

$$S = M^d \bmod n$$

4. Sends **Message (M) + Signature (S)**

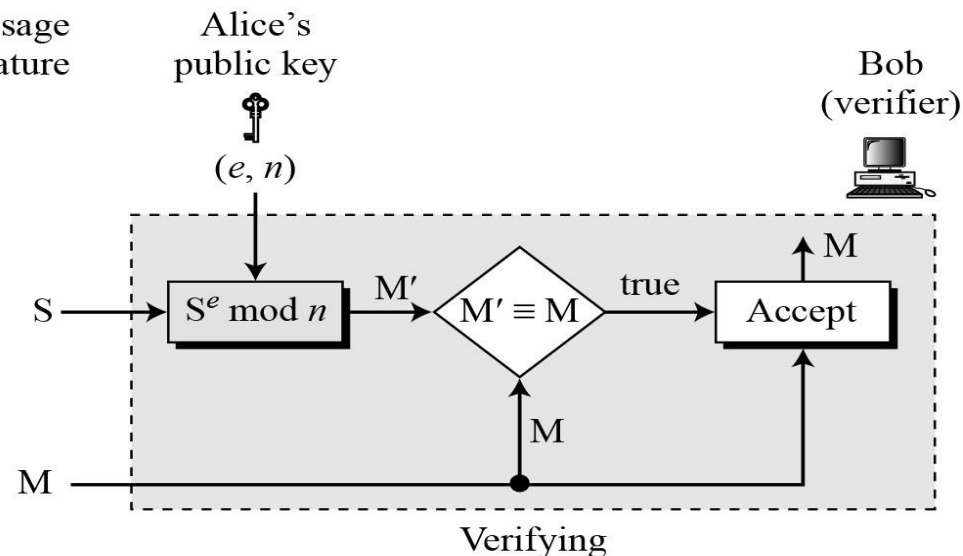


## Verification (Receiver – Bob)

1. Bob receives **M and S**
2. Uses **Alice's public key (e, n)**
3. Computes

$$M' = S^e \bmod n$$

4. If **M' = M** → **Accept**, otherwise **Reject**



# Example

As a trivial example, suppose that Alice chooses  $p = 823$  and  $q = 953$ , and calculates  $n = 784319$ . The value of  $\phi(n)$  is 782544. Now she chooses  $e = 313$  and calculates  $d = 160009$ . At this point key generation is complete. Now imagine that Alice wants to send a message with the value of  $M = 19070$  to Bob. She uses her private exponent, 160009, to sign the message:

## Phase 1: Key Generation (Alice's Setup)

Alice needs to create a public/private key pair.

**Choose Primes:** Alice selects two prime numbers.

$$p = 823$$

$$q = 953$$

**Compute  $n$  (Modulus):** She multiplies them to get  $n$ .

$$n = p \times q$$

$$n = 823 \times 953$$

$$n = 784319$$

**Compute  $\phi(n)$  (Euler's Totient):** This is needed to find the keys.

$$\phi(n) = (p - 1) \times (q - 1)$$

$$\phi(n) = (822) \times (952)$$

$$\phi(n) = 782544$$

**Choose  $e$  (Public Exponent):** She picks a number that is coprime to  $\phi(n)$ .

$$e = 313$$

**Compute  $d$  (Private Exponent):** She finds the number such that  $e \times d \equiv 1 \pmod{\phi(n)}$  (the modular inverse.)

$d = 160009$  (Given in the example)

*Check:*  $313 \times 160009 = 50082817$ . Divide by  $\phi(n)$ :  $50082817 \div 782544 = 64$  remainder 1. ✓

## Result of Key Generation:

**Public Key:** ( $n = 784319, e = 313$ ) <- Shared with everyone, especially Bob.

**Private Key:** ( $d = 160009$ ) <- Kept secret by Alice.

## Phase 2: Signing (Alice sends a message to Bob)

Alice wants to send the message  $M = 19070$ . She uses her **Private Key** to sign it.

**The Formula:**  $S = M^d \bmod n$

**The Calculation:**

$$S = 19070^{160009} \bmod 784319$$

*Note:* In real life, we don't compute this directly. We use fast exponentiation techniques.

**Given Result:** This simplifies to 210625.

**Result of Signing:**

**Message:**  $M = 19070$

**Signature:**  $S = 210625$

Alice sends the pair **(19070, 210625)** to Bob.

## Phase 3: Verification (Bob checks the signature)

Bob receives the pair **(19070, 210625)**.

He wants to verify it came from Alice.

He uses Alice's **Public Key** ( $n, e$ ).

**The Formula:**  $M' = S^e \bmod n$

**The Calculation:**

$$M' = 210625^{313} \bmod 784319$$

**Given Result:** This simplifies to 19070.

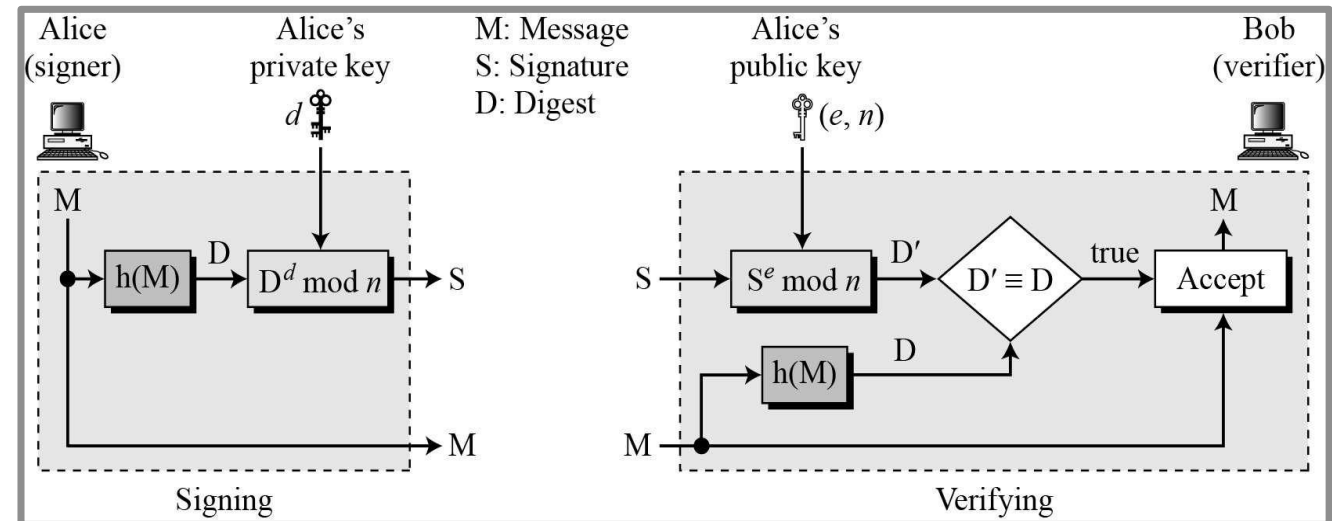
**The Comparison:**

Bob compares  $M'$  (the result of his calculation) with  $M$  (the message Alice sent).

$$M' = 19070$$

$$M = 19070$$

Since they match, the signature is **authentic**.



# RSA Signature Exercise



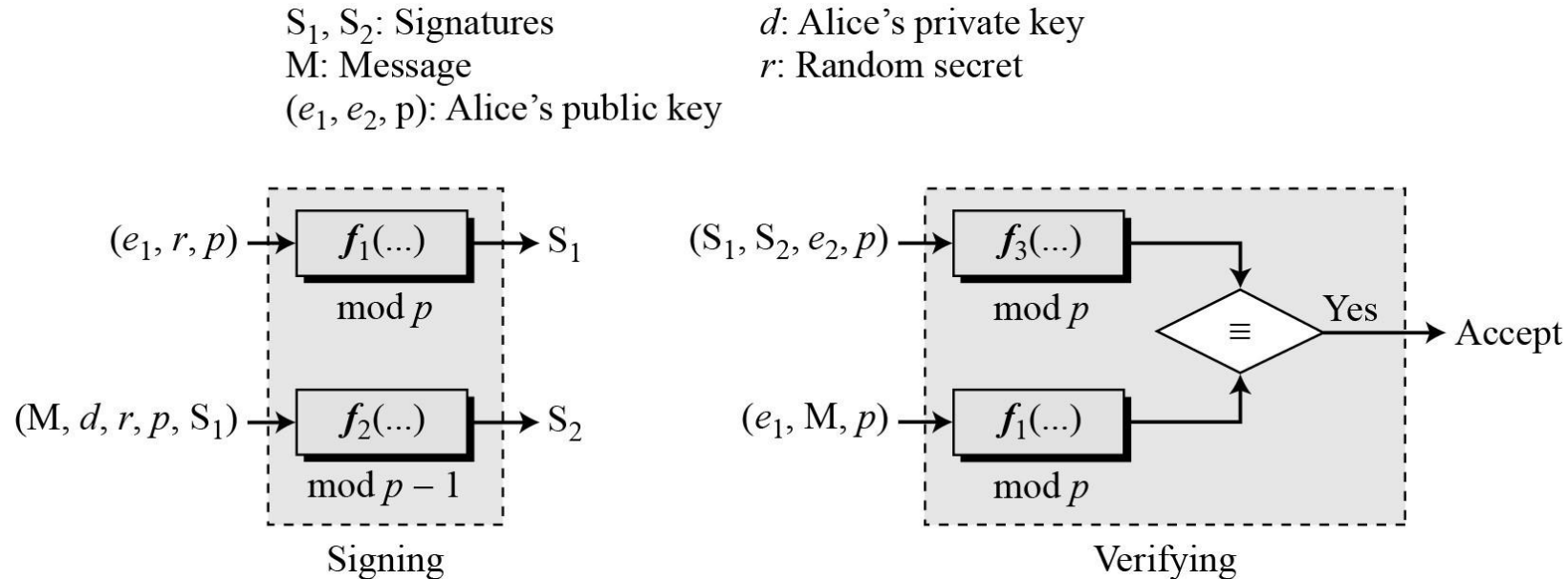
- Two prime numbers:  $p=7$ ,  $q=11$
- Public exponent:  $e=13$
- Message:  $M=9$



# RSA Signature on the Message Digest

- Apply RSA Digital Signature Scheme for signing and verifying the digest of the given message “**(3456709A)<sub>16</sub>**” with  **$p = 17$ ,  $q = 23$ , and  $e = 3$** . Message digest of **4 bits** is calculated as follows:
  - a. Arrange the bits of the message as **4 rows × 8 columns** matrix, in row order.
  - b. Apply **XOR operation row wise**
  - c. Take all the rows output bits (**4 bits**) as digest from **top to bottom** and **top is considered as MSB**.

# ElGamal Digital Signature Scheme



## Key Generation:

- The key generation procedure here is exactly the same as the one used in the cryptosystem.
- In ElGamal digital signature scheme,  **$(e_1, e_2, p)$**  is Alice's public key;  **$d$**  is her private key.



# Signing and Verifying

M: Message

$S_1, S_2$ : Signatures

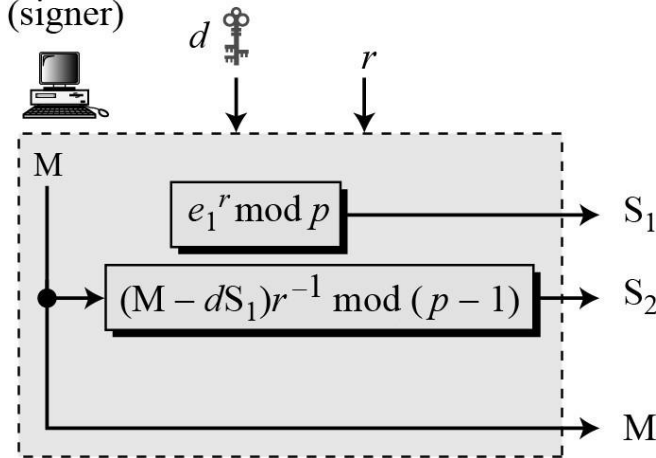
$V_1, V_2$ : Verifications

$r$ : Random secret

$d$ : Alice's private key

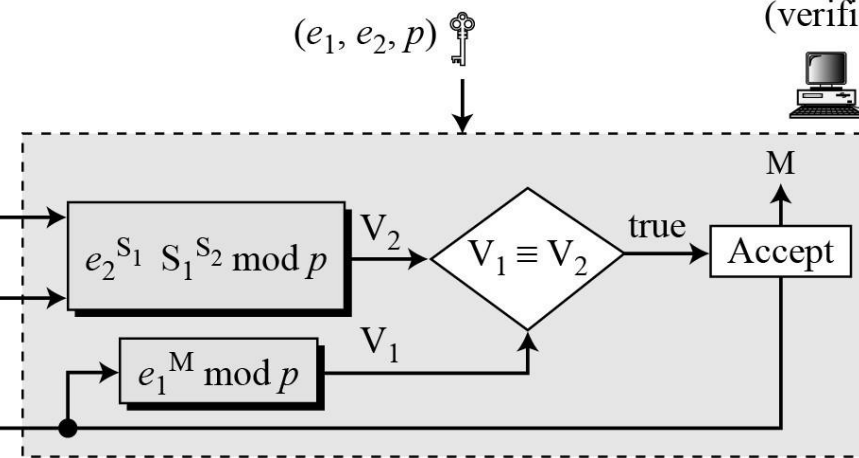
$(e_1, e_2, p)$ : Alice's public key

Alice  
(signer)



Signing

Bob  
(verifier)



Verifying

# ElGamal Digital Signature – Signing and Verification

## Signing Process (Alice)

1. Alice selects a random secret number  $r$  for each message
2. Computes first signature value  $S_1 = e_1^r \bmod p$
3. Computes second signature value
$$S_2 = (M - d \times S_1) \times r^{-1} \bmod (p - 1)$$
4. Sends **Message (M)** and **Signature (S<sub>1</sub>, S<sub>2</sub>)** to Bob

## Verification Process (Bob)

1. Check if  $0 < S_1 < p$  and  $0 < S_2 < p - 1$
2. Compute  $V_1 = e_1^M \bmod p$
3. Compute  $V_2 = e_2^{s_1} \times S_1^{s_2} \bmod p$
4. If  $V_1 = V_2 \rightarrow$  **Signature is valid**
5. Otherwise **Message is rejected**

# Example

Here is a trivial example. Alice chooses  $p = 3119$ ,  $e_1 = 2$ ,  $d = 127$  and calculates  $e_2 = 2^{127} \bmod 3119 = 1702$ . She also chooses  $r = 307$ . Alice announces  $e_1$ ,  $e_2$ , and  $p$  publicly; she keeps  $d$  secret. The following shows how Alice can sign a message.

## Step 1: Choose Public and Private Parameters

Alice selects some numbers required for the digital signature system.

$p = 3119$  (a large prime number)

$e_1 = 2$  (a generator)

$d = 127$  (Alice's **private key**)

Now Alice calculates her **public key value**:

$$e_2 = e_1^d \bmod p$$
$$e_2 = 2^{127} \bmod 3119 = 1702$$

### Explanation

**Public Key:**  $e_1, e_2, p \rightarrow$  Everyone can know these.

**Private Key:**  $d \rightarrow$  Only Alice knows this.

So Alice publishes:

$$e_1 = 2$$
$$e_2 = 1702$$
$$p = 3119$$

But keeps  $d = 127$  secret.

## PART 1: Alice Signs the First Message

### Step 2: Message to be Sent

Alice wants to send the message:

$$M = 320$$

To create a digital signature, she chooses a **random number**

$$r = 307$$

### Explanation

The value  $r$  **must be different for every message** to maintain security.

### Step 3: Calculate First Signature Value

$$S_1 = e_1^r \bmod p$$
$$S_1 = 2^{307} \bmod 3119 = 2083$$

### Explanation

This value becomes **the first part of the digital signature**.

#### Step 4: Calculate Second Signature Value

$$S_2 = (M - d \times S_1) \times r^{-1} \bmod (p - 1)$$

Substitute the values:

$$S_2 = (320 - 127 \times 2083) \times 307^{-1} \bmod 3118$$
$$S_2 = 2105$$

#### Explanation

Now Alice has two signature values:

$$S_1 = 2083$$

$$S_2 = 2105$$

These two numbers together form the **digital signature**.

#### Step 5: Alice Sends the Message

Alice sends the following to Bob:

Message: **M = 320**

Signature: (**S<sub>1</sub> = 2083** , **S<sub>2</sub> = 2105**)

## PART 2: Bob Verifies the Signature

Bob now checks if the message is really from Alice.

#### Step 6: Bob Calculates V<sub>1</sub>

$$V_1 = e_1^M \bmod p$$
$$V_1 = 2^{320} \bmod 3119 = 3006$$

#### Step 7: Bob Calculates V<sub>2</sub>

$$V_2 = e_2^{S_1} \times S_1^{S_2} \bmod p$$

Substitute values:

$$V_2 = 1702^{2083} \times 2083^{2105} \bmod 3119$$
$$V_2 = 3006$$

#### Step 8: Compare Results

Bob checks:

$$V_1 = V_2$$
$$3006 = 3006$$

Since both values are equal,

✓ **The signature is valid.**

✓ **Bob accepts the message.**

#### Second Message Example

Alice now sends another message to **Ted**.

#### Step 1: New Message

$$M = 3000$$

Alice must choose a **new random number**

$$r = 107$$

#### Step 2: Compute First Signature

$$S_1 = e_1^r \bmod p$$
$$S_1 = 2^{107} \bmod 3119 = 2732$$

#### Step 3: Compute Second Signature

$$S_2 = (M - d \times S_1) \times r^{-1} \bmod (p - 1)$$
$$S_2 = (3000 - 127 \times 2732) \times 107^{-1} \bmod 3118$$
$$S_2 = 2526$$

#### Step 4: Ted Verifies the Message

Ted calculates:

##### First Value

$$V_1 = e_1^M \bmod p$$
$$V_1 = 2^{3000} \bmod 3119 = 704$$



## Second Value

$$V_2 = e_2^{s_1} \times S_1^{s_2} \bmod p$$
$$V_2 = 1702^{2732} \times 2732^{2526} \bmod 3119$$
$$V_2 = 704$$

## Step 5: Final Verification

$$V_1 = V_2$$
$$704 = 704$$

- ✓ Signature is **valid**
- ✓ Ted accepts the message.



# Elgamal Digital Signature Example

Prime:  $p=23$

$e=5$

Private key:  $d=6$

Message:  $M=13$

$r=7$

# Digital Signature Standard (DSS)

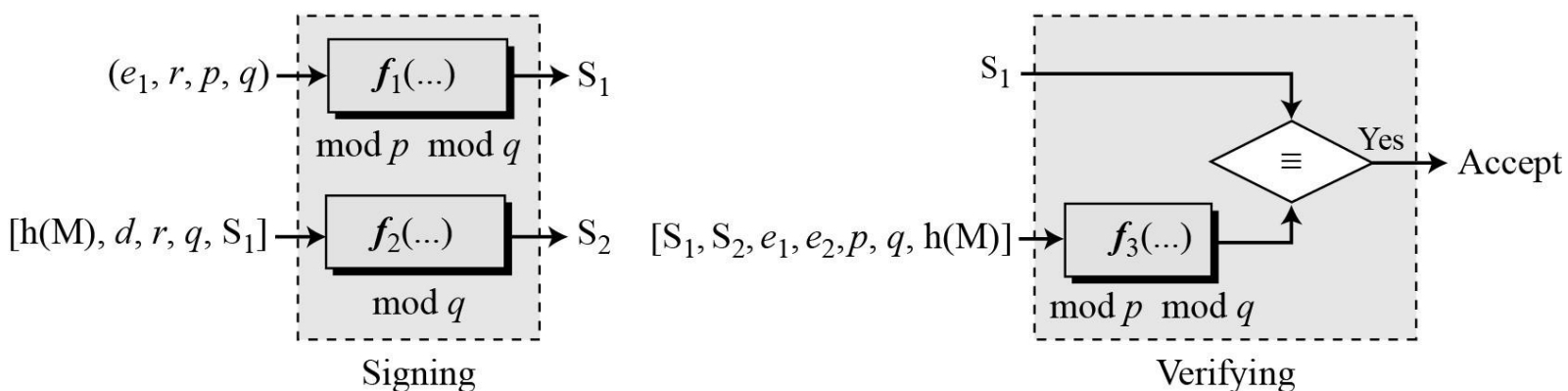
$S_1, S_2$ : Signatures

$M$ : Message

$(e_1, e_2, p, q)$ : Alice's public key

$d$ : Alice's private key

$r$ : Random secret



## Key Generation

1. Alice selects two prime numbers **p** and **q**
2. Generates value **e<sub>1</sub>**
3. Chooses secret key **d**
4. Computes **e<sub>2</sub> = e<sub>1</sub><sup>d</sup> mod p**
5. **Public Key:**  $(e_1, e_2, p, q)$
6. **Private Key:** **d**

## Signing Process

1. Alice selects random number **r**
2. Computes message hash **h(M)**
3. Generates two signature values **S<sub>1</sub>** and **S<sub>2</sub>**
4. Sends **Message (M) + Signature (S<sub>1</sub>, S<sub>2</sub>)**

## Verification

1. Receiver uses **public key**
2. Calculates verification value
3. If calculated value = **S<sub>1</sub>** → **Accept message**
4. Otherwise → **Reject message**

# Key Generation

Before sending a digitally signed message, Alice must create **public and private keys**.

## Steps:

### 1. Choose a prime number ( $p$ ):

Alice selects a large prime number  $p$  (512–1024 bits).

### 2. Choose another prime number ( $q$ ):

Alice selects a **160-bit prime  $q$**  such that  **$q$  divides  $(p - 1)$** .

### 3. Create a generator value ( $e_1$ ):

Using mathematical operations, Alice generates a value  $e_1$  which will be used in the signature process.

### 4. Select private key:

Alice chooses a secret number  $d$ .

This number is her **private key** and must be kept secret.

### 5. Compute public key value:

Alice calculates  $e_2 = e_1^d \bmod p$ .

### 6. Final Keys:

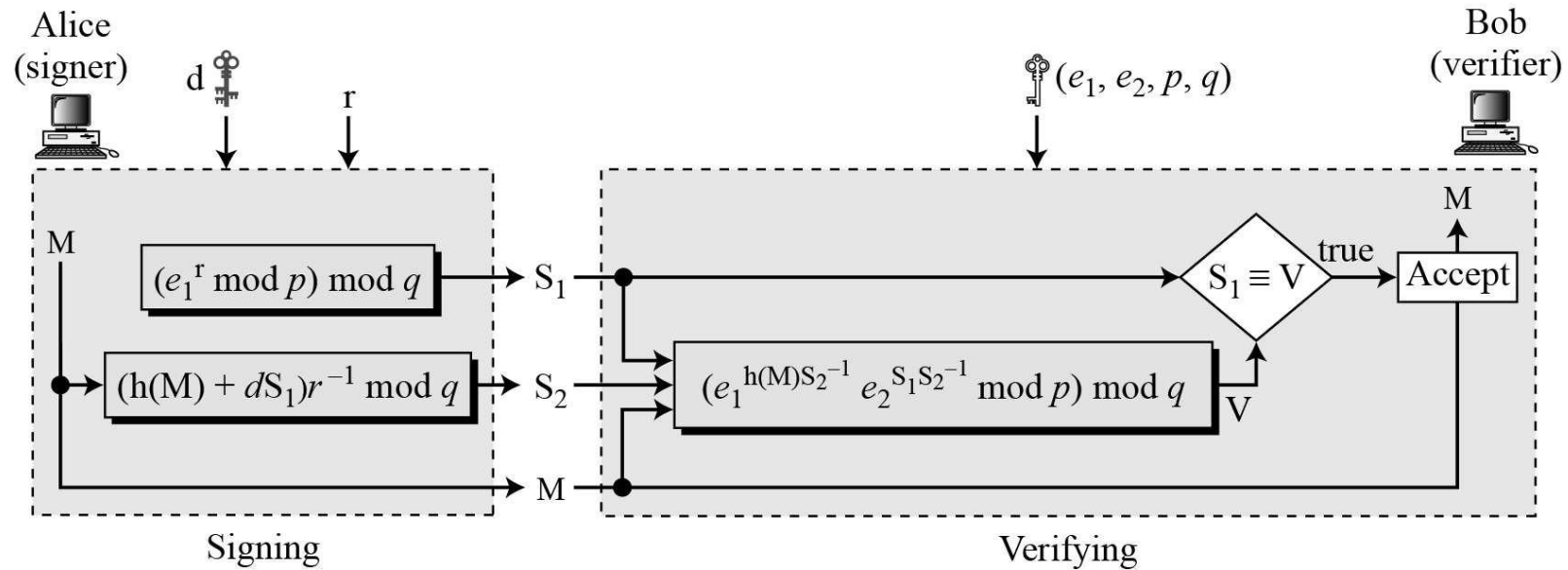
**Public Key:**  $(e_1, e_2, p, q) \rightarrow$  Shared with everyone

**Private Key:**  $d \rightarrow$  Kept secret by Alice



# Digital Signature Standard (DSS)

M: Message       $r$ : Random secret       $h(M)$ : Message digest  
 $S_1, S_2$ : Signatures       $d$ : Alice's private key  
V: Verification       $(e_1, e_2, p, q)$ : Alice's public key





## Signing the Message (Done by Alice)

1. Alice chooses a **random number  $r$**  between **1 and  $q$** .  
A new  **$r$  must be selected every time a new message is signed**.
2. Alice calculates the **first signature value  $S_1$**  using the formula:
$$S_1 = (e_1^r \bmod p) \bmod q$$
3. Alice creates a **hash value of the message**, written as  **$h(M)$** .  
A hash converts the message into a fixed-size number.
4. Alice calculates the **second signature value  $S_2$**  using:
$$S_2 = (h(M) + dS_1)r^{-1} \bmod q$$
5. Alice sends **three things to Bob**:  
Message  **$M$**   
Signature values  **$S_1$  and  $S_2$**
6. These two values ( **$S_1, S_2$** ) form the **digital signature**.

## Verifying the Signature (Done by Bob)

When Bob receives the message, he verifies the signature.

1. Bob checks if  **$S_1$  and  $S_2$  are within valid ranges**:
$$0 < S_1 < q$$
$$0 < S_2 < q$$
2. Bob calculates the **hash of the received message** using the same hash function used by Alice.
3. Bob calculates a verification value  **$V$**  using the public key values.
4. Bob compares  **$S_1$  and  $V$** .  
If  **$S_1 = V \rightarrow$  Message is accepted**  
If  **$S_1 \neq V \rightarrow$  Message is rejected**

This ensures the message is **authentic and unchanged**.

# Example

Alice chooses  $q = 101$  and  $p = 8081$ . Alice selects  $e_0 = 3$  and calculates  $e_1 = e_0^{(p-1)/q} \bmod p$ . Alice chooses  $d = 61$  as the private key and calculates  $e_2 = e_1^d \bmod p$ . Now Alice can send a message to Bob. Assume that  $h(M) = 5000$  and Alice chooses  $r = 61$ .

## Step 1: Given Values

$$\begin{aligned}q &= 101 \\p &= 8081 \\e_0 &= 3\end{aligned}$$

## Step 2: Calculate $e_1$

$$e_1 = e_0^{(p-1)/q} \bmod p$$

First compute:

$$\begin{aligned}\frac{p-1}{q} &= \frac{8081-1}{101} \\&= \frac{8080}{101} = 80\end{aligned}$$

Now

$$\begin{aligned}e_1 &= 3^{80} \bmod 8081 \\e_1 &= 6968\end{aligned}$$

## Step 3: Private Key

Given

$$d = 61$$

## Step 4: Calculate $e_2$

$$\begin{aligned}e_2 &= e_1^d \bmod p \\e_2 &= 6968^{61} \bmod 8081 \\e_2 &= 2038\end{aligned}$$

Public key =  $(e_0, e_1, e_2, p, q)$

Private key =  $d = 61$

## Step 5: Message Hash

$$h(M) = 5000$$

Random number chosen

$$r = 61$$

## Step 6: Calculate $S_1$

$$\begin{aligned}S_1 &= (e_1^r \bmod p) \bmod q \\S_1 &= (6968^{61} \bmod 8081) \bmod 101 \\S_1 &= 54\end{aligned}$$

### Step 7: Calculate $S_2$

Formula

$$S_2 = ((h(M) + dS_1)r^{-1}) \bmod q$$

Substitute values

$$S_2 = ((5000 + 61 \times 54)r^{-1}) \bmod 101$$

First calculate

$$61 \times 54 = 3294$$

$$5000 + 3294 = 8294$$

Now compute  $r^{-1} \bmod 101$

$$r = 61$$

Multiplicative inverse:

$$\begin{aligned} 61^{-1} \bmod 101 \\ = 58 \end{aligned}$$

Now

$$\begin{aligned} S_2 &= (8294 \times 58) \bmod 101 \\ S_2 &= 40 \end{aligned}$$

### Step 8: Signature Sent to Bob

Alice sends

$$\begin{aligned} (M, S_1, S_2) \\ S_1 &= 54 \\ S_2 &= 40 \end{aligned}$$

### Step 9: Verification by Bob

First compute

$$\begin{aligned} S_2^{-1} \bmod q \\ 40^{-1} \bmod 101 = 48 \end{aligned}$$

### Step 10: Compute $V$

$$V = [(6968^{5000 \times 48} \times 2038^{54 \times 48}) \bmod 8081] \bmod 101$$

After modular calculation

$$V = 54$$

### Step 11: Final Verification

$$\begin{aligned} V &= S_1 \\ 54 &= 54 \end{aligned}$$

✅ **Signature is valid.**



# DSS Exercise

$p=23, q=11$

$e_1=2$

Private key  $d=3$

$h(M)=9$

Random  $r=4$